

DecisionIQ Master Handbook

CHAPTER 9: The ETL Ingestion Inbound Pipeline and Quality Audits

9.1 The ETL Architecture

The ETL pipeline extracts raw operational data, validates constraints, normalizes columns, and bulk-loads database tables:

```
[Raw CSVs] → [Clean & Normalize] → [Range & FK Validation] → [SQLAlchemy Bulk Load]
```

9.2 Ingestion Steps

1. **Extract:** Ingestion scripts read raw CSVs into memory using Pandas.
2. **Transform:**
 - Whitespace is trimmed from text columns.
 - Categories are converted to uppercase.
 - Dates are normalized to `YYYY-MM-DD` format.
3. **Validate:** The data is routed through a series of programmatic checks before loading.
4. **Load:** Tables are loaded using SQLAlchemy bulk transactions.

9.3 Inbound Validation Rules

The ETL validation engine (`validator.py`) enforces three categories of checks:

1. Primary Key Audits

Ensures that the primary key for each table is unique and non-null:

```
if df[pk_col].isnull().sum() > 0 or df[pk_col].duplicated().sum() > 0:  
    raise ValueError(f"PK audit failed for table: {table_name}")
```

2. Range Audits

Ensures that prices, costs, and order quantities are positive values:

```
if (df[col] < min_val).sum() > 0:  
    raise ValueError(f"Range violation in column {col}")
```

3. Foreign Key Audits

Verifies that all child table foreign keys exist in the parent tables, maintaining perfect referential integrity.